

Sum of Nat
arith
geo

EO16

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Assignment 4

$$\begin{aligned}
 1. \quad & x \quad 2R \rightarrow 40 \quad 2W \rightarrow 20 \quad 1R, 1W \rightarrow 30 \\
 & P(X=x) \quad \frac{{}^3C_2 \times {}^2C_0}{{}^5C_2} \quad \frac{{}^2C_2 \times {}^3C_0}{{}^5C_2} \quad \frac{{}^3C_1 \times {}^2C_1}{{}^5C_2} \\
 & = \frac{3}{10} \quad = \frac{1}{10} \quad = \frac{3}{5}
 \end{aligned}$$

$$\sum x_i p_i \quad 12 \quad 2 \quad 12$$

$$\boxed{\sum x_i p_i = E(X) = 32}$$

$$\begin{aligned}
 2. \quad & x \quad 1 \quad 2 \quad 3 \quad \dots \quad n \\
 & P(X=x) \quad \frac{1}{n} \quad \frac{1}{n} \quad \frac{1}{n} \quad \dots \quad \frac{1}{n}
 \end{aligned}$$

$$\sum p_i x_i = \frac{1}{n}(1) + \frac{1}{n}(2) + \frac{1}{n}(3) + \dots + \frac{1}{n}(n)$$

Series for Sum of Natural Numbers

$$\boxed{E(X) = \frac{n+1}{2}}$$

$$\begin{aligned}
 \text{Var}(X) &= E(X^2) - [E(X)]^2 \\
 &= \frac{1}{n}(1^2) + \frac{1}{n}(2^2) + \frac{1}{n}(3^2) + \dots + \frac{1}{n}(n^2)
 \end{aligned}$$

$$= \frac{1}{n} \left[\frac{n(n+1)(2n+1)}{6} \right] = \frac{(n+1)(2n+1)}{6}$$

$$\text{Var}(X) = \frac{2n^2 + n + 2n + 1}{6} - \frac{(n+1)^2}{4}$$

$$= \frac{2n^2 + 3n + 1}{6} - \frac{3n + 3}{6}$$

$$= \frac{2n^2 + 3n + 1}{6} - \frac{n^2 + 1 + 2n}{4}$$

$$= \frac{8n^2 + 12n + 4 - 6n^2 - 6 - 12n}{24}$$

$$= \frac{2n^2 - 2}{24} = \frac{n^2 - 1}{12}$$

$$\boxed{\text{Var}(X) = \frac{n^2 - 1}{12}}$$

4.	X_i	0	1	2	3
	P_i	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$
	$P_i X_i$	0	$\frac{3}{8}$	$\frac{6}{8}$	$\frac{3}{8}$

$$E(X) = \sum P_i X_i = \frac{12}{8} = \boxed{1.5}$$

$$\text{Var}(X) = E(X^2) - (E(X))^2$$

$$E(X^2) = \frac{1}{8}(0^2) + \frac{3}{8}(1^2) + \frac{3}{8}(2^2) + \frac{1}{8}(3^2)$$

$$= \frac{3}{8} + \frac{3}{2} + \frac{9}{8} = 3$$

$$[E(X)]^2 = (1.5)^2 = 2.25$$

$$\text{Var}(X) = 3 - 2.25 = \frac{3}{4} = \boxed{0.75}$$

5. $f(x) = K(1+x)$, $2 \leq x \leq 5$

(i) $K = ?$, (ii) $P(X \leq 4)$, (iii) $E(X)$

(i) PDF $\rightarrow \int_{-\infty}^{\infty} f(x) dx = 1$

$$K \int_2^5 (1+x) dx = 1 \rightarrow K \left[x + \frac{x^2}{2} \right]_2^5 = 1$$

$$= K \left[5 + \frac{25}{2} - 2 - \frac{4}{2} \right] = 1$$

$$K = \frac{2}{27}$$

(ii) $P(X \leq 4)$

$$\frac{2}{27} \int_2^4 (1+x) dx \rightarrow \frac{2}{27} \left[x + \frac{x^2}{2} \right]_2^4$$

$$= \frac{2}{27} \left[4 + \frac{16}{2} - 2 - \frac{4}{2} \right] = \frac{2}{27} [2 + 6]$$

$$= \frac{16}{27} \quad (ii) = \boxed{0.59}$$

(iii) $E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$

$$= \frac{2}{27} \int_2^5 x + x^2 dx \rightarrow \frac{2}{27} \left[\frac{x^2}{2} + \frac{x^3}{3} \right]_2^5$$

$$= \frac{2}{27} \left[\frac{25}{2} + \frac{125}{3} - \frac{4}{2} - \frac{8}{3} \right]$$

$$= \frac{2}{27} \left[\frac{21}{2} + \frac{117}{3} \right] = \frac{11}{3} = \boxed{3.67}$$

6.

X	1	3	4	5
P(X=x)	0.4	0.1	0.2	0.3
Pi xi	0.4	0.3	0.8	1.5
E(X) = $\sum p_i x_i$	= $\boxed{3}$			

$$\text{Var}(X) = E(X^2) - [E(X)]^2$$

$$E(X^2) = 0.4 + (0.1 \times 9) + (0.2 \times 16) + (0.3 \times 25) = 12$$

$$[E(X)]^2 = 9$$

$$\text{Var}(X) = 12 - 9 = \boxed{3}$$

7. $f(x) = \begin{cases} kx & , 0 \leq x \leq 2 \\ 2k & , 2 \leq x \leq 4 \\ 6k - kx & , 4 \leq x \leq 6 \end{cases}$

(i) $k = ?$, (ii) $P(1 \leq X \leq 3)$, (iii) $E(X)$

(i) $\int_0^2 kx dx + \int_2^4 2k dx + \int_4^6 (6k - kx) dx = 1$

$$k \left[\frac{x^2}{2} \right]_0^2 + 2k \left[x \right]_2^4 + \left[6kx - \frac{kx^2}{2} \right]_4^6 = 1$$

$$= K \left[\frac{4}{2} \right] + 2K [4-2] + \left[36K - \frac{36}{2}K - 24K + \frac{16K}{2} \right]$$

$$= 2K + 4K + 18K - \frac{20K}{2} = 1$$

$$= 18K - 10K = 1 \rightarrow 8K = 1 \rightarrow \boxed{K = \frac{1}{8}}$$

$$(ii) P(1 \leq X \leq 3) = \frac{1}{8} \int_1^2 x \cdot dx + \frac{1}{4} \int_2^3 1 \cdot dx$$

$$= \frac{1}{8} \left[\frac{x^2}{2} \right]_1^2 + \frac{1}{4} [x]_2^3$$

$$= \frac{1}{8} \left[\frac{4}{2} - \frac{1}{2} \right] + \frac{1}{4} [3-2]$$

$$P(1 \leq X \leq 3) = \frac{1}{8} \left[\frac{3}{2} \right] + \frac{1}{4} = \frac{3}{16} + \frac{1}{4} = \boxed{\frac{7}{16}}$$

$$(iii) E(X) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$E(X) = \frac{1}{8} \int_0^2 x^2 dx + \frac{2}{8} \int_2^4 x dx + \int_4^6 \left[\frac{6x}{8} - \frac{x^2}{8} \right] dx$$

$$= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^2 + \frac{2}{8} \left[\frac{x^2}{2} \right]_2^4 + \left[\frac{6x^2}{16} - \frac{x^3}{24} \right]_4^6$$

$$= \frac{1}{8} \left[\frac{8}{3} \right] + \frac{2}{8} \left[\frac{16}{2} - \frac{4}{2} \right] + \left[\frac{6 \cdot 36}{16} - \frac{216}{24} - 6 + \frac{64}{24} \right]$$

$$E(X) = \frac{1}{3} + \frac{3}{2} + \frac{7}{6} = \boxed{3}$$

$$8. f(x) = \begin{cases} 2e^{-2x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

$$(i) E(X) = 2 \int_0^{\infty} x e^{-2x} dx$$

$$E(X) = 2 \left[\frac{x \cdot e^{-2x}}{-2} - \int \frac{1 \cdot e^{-2x}}{-2} dx \right]_0^{\infty}$$

$$= 2 \left[\frac{x e^{-2x}}{-2} + \frac{1}{2} \left[\frac{e^{-2x}}{-2} \right] \right]_0^{\infty}$$

$$= 2 \left[\frac{x \cdot e^{-2x}}{-2} - \frac{1}{4} e^{-2x} \right]_0^{\infty}$$

$$= 2 \left[\frac{1}{4} \right] = \frac{1}{2}$$

$$(ii) E(X^2) = 2 \int_0^{\infty} x^2 \cdot e^{-2x} dx$$

$$= 2 \left[\frac{x^2 \cdot e^{-2x}}{-2} - \int \frac{2x \cdot e^{-2x}}{-2} dx \right]_0^{\infty}$$

$$= 2 \left[\frac{-x^2 e^{-2x}}{2} + \left[\frac{x \cdot e^{-2x}}{-2} - \int \frac{e^{-2x}}{-2} dx \right] \right]_0^{\infty}$$

$$= 2 \left[\frac{-x^2 e^{-2x}}{2} + \left[\frac{x e^{-2x}}{-2} + \frac{1}{2} \left[\frac{e^{-2x}}{-2} \right] \right] \right]_0^{\infty}$$

$$= 2 \left[\frac{-x^2 e^{-2x}}{2} + \left[\frac{x e^{-2x}}{-2} - \frac{e^{-2x}}{4} \right] \right]_0^{\infty}$$

$$= 2 \left[\frac{-x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} \right]_0^\infty$$

$$= 2 \left[\frac{1}{4} \right] = \boxed{\frac{1}{2}}$$

(iii) $\text{Var}(X) = E(X^2) - [E(X)]^2$

$$= \frac{1}{2} - \frac{1}{4} = \boxed{\frac{1}{4}}$$

(iv) S.D of X = $\sqrt{\frac{1}{4}} = \boxed{\frac{1}{2}}$

9. $f(x) = \begin{cases} 2ax + b, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$

$$E(X) = 3$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$\int_0^2 x(2ax + b) dx = 3$$

$$\left[\frac{2ax^3}{3} + \frac{bx^2}{2} \right]_0^2 = 3$$

$$\frac{16a}{3} + 2b = 3$$

$$16 + 6b = 9 \rightarrow 16a + 6b - 9 = 0$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\left[\frac{2ax^2 + bx}{2} \right]_0^2 = 1$$

$$4a + 2b = 1 \rightarrow \boxed{a = \frac{3}{2}, b = -\frac{5}{2}}$$

3. Event	H	TH	TTH	TTTH
X	1	2	3	4
$P(X=x)$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$

$$E(X) = \sum p_i x_i$$

$$S = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{16}(4) + \dots$$

$$\frac{1}{2S} = \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots$$

$$S - \frac{1}{2S} = \frac{1}{2} \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \right]$$

$$= \frac{1}{2} \left[\frac{1}{1 - \frac{1}{2}} \right] = 1$$

$$S = E(X) = 1 + 1 = 2$$

Entry Fee = Rs. 2

$$\text{Net gain} = 2 - 2 = \boxed{0}$$

~~RVP~~
22/10/21